

NEYL GASMI

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Quantitative Research | M2MO, Université Paris Cité/ENSAE | Class of 2027

Education

Université Paris Cité <i>Master's degree in Probability and Finance, M2MO</i>	2026 – 2027 <i>Paris, France</i>
• Relevant Coursework: Stochastic Optimal Control, Algorithmic Trading (O. Guéant), Deep Learning, Reinforcement Learning, Optimal Transport, Derivative Modeling, C++.	
Université Paris 1 Panthéon-Sorbonne <i>M1, Mathematics applied to finance; ranked 1st/45 in Stochastic Processes</i>	2025 – 2026 <i>Paris, France</i>
Université Paris Dauphine-PSL <i>Bachelor in Mathematics & Physics</i>	2025 – 2026 <i>Paris, France</i>
Université de Lille <i>Bachelor in Mathematics & Statistics</i>	2024 – 2025 <i>Lille, France</i>
EDHEC Business School <i>Grande École Program, MiM</i>	2024 – 2027 <i>Lille, France</i>

Experience

EuroCTP (ESMA-selected EU Consolidated Tape Provider) <i>Quantitative Research Intern, Equities & ETFs</i>	Summers 2025 & 2026 <i>Amsterdam, Netherlands</i>
• Reconstructed the European consolidated tape from Level 3 (full order book) BMLL data across 30+ venues (Euronext, CBOE Europe, Deutsche Börse, Aquis, Nasdaq Nordic) and 10^5 instruments; engineered EBBO, mid-price and bid-ask spread dynamics from asynchronous, microsecond-timestamped multi-venue streams. • Quantified how cross-venue clock desynchronisation biases spread and realised volatility estimates, and how naive aggregation misstates liquidity in fragmented markets, informing the tape's core methodology. • Sole author of the research report formalising the signal-coherence evaluation methodology for the consolidated tape, delivered to ESMA.	
Laboratoire SAMM, Université Paris 1 <i>Research Intern, Supervised by Alain Celisse</i>	March 2026 – Sept 2026 <i>Paris, France</i>
• Designed a Mamba (selective SSM) encoder for classification of irregularly sampled, noisy time series (OGLE-IV astronomical survey), operating on raw timestamps without resampling, phase-folding or period estimation. • Constructed a geometrically structured latent space via DTW-informed triplet loss with ordinal constraints; derived Lipschitz bounds on the selective state space to guarantee representation stability. • Benchmarked against Transformer and LSTM baselines at iso-parameter count; diagnosed representational collapse under class imbalance via alignment/uniformity metrics.	

Projects

Online Jump Detection with Provable FDR Guarantees <i>Python</i> 	2025 – 2026
• Built an online jump detection library (86 unit tests) for tick-by-tick price processes, combining composite e-values (GROW framework) with e-LOND/e-BH sequential procedures to control FDR across thousands of simultaneous hypothesis tests. • Exposed a failure mode of the standard Benjamini–Hochberg procedure in the high-frequency regime: 10,000 Monte Carlo replications (Merton, Heston, SVJ) show BH violating FDR at 5-second sampling (9.6% vs. 5% target) from bipower-variation bias under serial dependence, while e-value procedures remain valid. • Mapped the power/FDR frontier against Lee–Mykland benchmarks across sampling frequencies (1s–1min) and jump intensities, identifying where classical detection breaks down at high frequency.	

Technical Skills

Languages: Python, R, SQL, $\text{\LaTeX}$, C++

Libraries / Frameworks: PyTorch, PySpark, NumPy, Pandas, Scikit-learn

Tools: GitHub, VS Code, AWS Athena, DataDog, TimeBase, BMLL, Corvil

Certifications: TOEIC

Languages (spoken): French (native), English (C1)